SPA2D: Shift-Plus-Add for Autonomous Battery Unit Operations via Detect-and-Track of Tiny, Long-Distance Aerial Targets

***Abstract*—Long-range autonomous aerial rendezvous requires detection and tracking of targets occupying only a few pixels in onboard images. This challenge is a key aspect of Autonomous Battery Unit (ABU) operations, where a detachable battery system visually acquires distant host aircraft long before close-range navigation and docking. At such distances, target electric, vertical-takeoff-and-landing (eVTOL) aircraft exhibit low signal-to-noise ratio (SNR), weak spatial structure, and motion that is difficult to distinguish from clutter or sensor noise. This work proposes Shift-Plus-Add (SPA2D), a track-before-detect framework for low-pixel-count aerial target detection and tracking. SPA2D evaluates candidate image-plane motion hypotheses, aligns consecutive frames according to each hypothesis, and accumulates the aligned image sequences so that information consistent with target motion adds coherently while suppressing uncorrelated background information. To make this search practical, SPA2D combines GPU-accelerated trajectory evaluation with dynamic local tracking, which restricts later searches around the previously estimated target state. Experiments on two ABU-like sky-tracking sequences and two stress tests (faster motion, more clutter, and higher resolution) show that SPA2D achieves more than 97% visible-target recall and 98% precision on the ABU-like cases, and more than 84% recall and 93% precision on the stress tests. On embedded Jetson AGX Orin hardware, ABU-like test cases achieve between 16 and 28 frames per second (FPS). These results support motion-compensated temporal accumulation (i.e., shift-plus-add) for long-range aerial perception.**

***Index Terms*—vision-based tracking, aerial target detection, autonomous aerial systems, shift-plus-add, urban air mobility**

I. Introduction

Advanced air mobility (AAM) concepts such as eVTOL aircraft promise quiet, economical flight after addressing challenges of operational cadence due to battery energy density, charging time, and onboard battery mass. ABUs [1] propose a system-level solution to these challenges: detachable battery systems autonomously rendezvous with, attach to, and detach from a host aircraft during flight to provide supplemental energy. This novel architecture requires autonomy: before close-range relative navigation and docking occurs, the ABU must visually detect and track a distant host aircraft from onboard images. Long range targets occupy only a few pixels with low SNR and may move against background clutter.

Vision-based airborne object detection spans image-processing sense-and-avoid methods [2], airborne collision detection [3], and broader reviews of autonomous vision-based avoidance [4]. Prior work addresses passive monocular ranging [5], below-horizon aircraft detection [6], [7], learning-based detectors [8], and onboard tracking [9], [10]. Related drone work includes cooperative unmanned aerial vehicle (UAV) tracking [11], vision-based anti-UAV detection and tracking [12], and broader UAV surveys [13]. Despite these efforts, long-range ABU-like rendezvous remains challenging because the target is so small that it lacks stable features across frames.

Recent object detectors include region-based convolutional detectors [14], real-time one-stage detection [15], feature pyramids [16], and transformer detectors [17], [18]. Small-target methods use cross-scale detection transformers [19], cross-ensemble feature fusion [20], scale- and location-sensitive infrared models [21], dense nested attention [22], and attentional local contrast [23]; single-frame infrared dim-small-target methods are reviewed in [24]. These detection-first approaches depend on sufficient spatial evidence within each frame. Aerial targets appearing as a few, dim pixels can be difficult to distinguish from noise under these techniques.

This work proposes Shift-Plus-Add (SPA2D), a track-before-detect framework for low-pixel-count aerial target detection and tracking. Rather than detecting the target independently in each image, SPA2D evaluates candidate image-plane motion hypotheses over a short time window. For each hypothesis, SPA2D shifts consecutive frames to compensate for the hypothesized target motion and then accumulates. A correct trajectory hypothesis causes target information to accumulate coherently into a stronger response, and an incorrect hypothesis smears target information across uncorrelated background information. SPA2D uses motion consistency to repair weak target signatures that cannot be detected in individual frames.

A direct search over image-plane motion hypotheses can be computationally expensive, especially for large image sizes and wide velocity ranges. To address this issue, SPA2D combines GPU-accelerated trajectory testing with localized searches. The GPU acceleration exploits the parallel structure of the hypothesis search, while localized searches only occur around previously estimated target locations.

In summary, this work makes the following contributions:

1) A Shift-Plus-Add (SPA2D) track-before-detect framework for long-range aerial target detection and tracking, designed for low-SNR, low-pixel-count targets.
2) A parallelized motion-compensated temporal accumulation trajectory search for recovering target position and motion from dim per-frame data. Coarse-to-fine search correctly recovers motion at 40 FPS on the Jetson GPU compared to 6 FPS on CPU.
3) A dynamic local tracking system combining global acquisition and local refinement with GPU acceleration. On the ABU-like sky-tracking sequences, SPA2D achieves more than 97% visible-target recall and 98% precision at 16 to 28 FPS on the Jetson AGX Orin and maintains more than 84% recall and 93% precision on stress tests.

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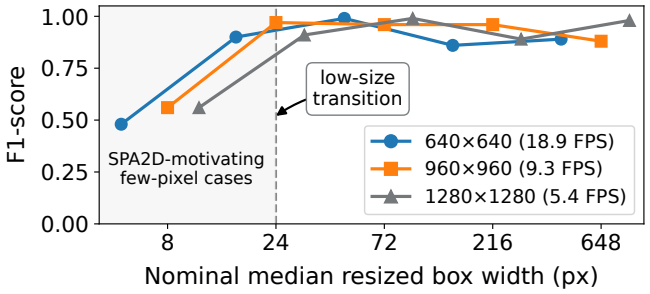


Fig. 1. CNN object detection performs well until target pixel counts fall below an empirical threshold. Object detection accuracy drops significantly for small pixel counts, making CNNs unreliable for the ABU application.

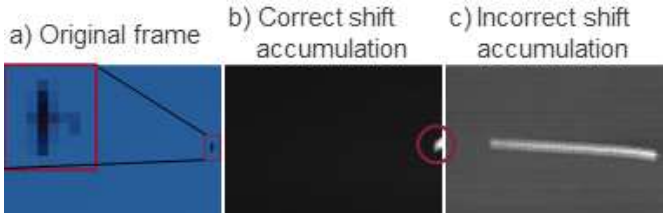


Fig. 2. Left: A target aircraft occupies very few pixels. Middle: When the velocity hypothesis is correct, target pixels accumulate coherently. Right: When the velocity hypothesis is incorrect, target pixels do not accumulate.

II. Background

This work addresses a key challenge in Autonomous Battery Unit (ABU) operations: autonomous aerial rendezvous with long-range, dimly-visible eVTOLs. Urban air mobility (UAM) for eVTOL aircraft struggles against constraints of battery energy density, which remains roughly $50\times$ lower than aviation gasoline by mass with little improvement in commercial systems over the past decade. Recent work [1] addresses this challenge at the system level with ABUs consisting of one or more rotors, batteries, and an attitude determination and control system (ADACS) with onboard sensors and compute that attaches to and detaches from a host aircraft during flight. This approach mirrors aerial refueling [25] and fuel drop tanks used by military aircraft while introducing emerging autonomous navigation capabilities. ABUs support assisted takeoff to offset an energy-intensive portion of flight; in-flight detachment to shed depleted battery mass; in-flight rendezvous to replenish energy reserves; and assisted landing for safety during divert scenarios. Inspired by ABUs, this work advances long-range detect-and-track for autonomous rendezvous.

Our shift-plus-add framework draws on a history of image accumulation methods in astronomy, which enhances weak visual signals by intelligently accumulating multiple observations. For example, astronomical speckle imaging aligns and accumulates short-exposure images distorted by atmospheric turbulence to recover higher-resolution images [26], [27]. Subsequent works introduced self-calibration [28] and zero-and-add [29] to reduce bias and artifacts, with practical use demonstrated in high-spatial-resolution telescope imaging [30]. Related approaches in moving-object astronomy shift sequential images according to a motion hypothesis to coherently accumulate observations of faint asteroids while suppressing background stars and noise [31]–[33]. SPA2D employs shift-plus-add concepts for aerial detect-and-track at distances much closer than these astronomical scales.

III. Motivation

We motivate SPA2D for ABUs with two studies. First, we quantify the impact of target pixel count on state-of-the-art (SotA) object detection, showing that CNNs accuracy degrades sharply for long-range, dimly-visible targets. Second, we apply shift-plus-add to synthetic data from a custom simulator for a performance baseline under this alternative detect-and-track method, showing that shift-plus-add succeeds where CNNs fail. These studies provide insights into different shift-plus-add algorithm designs, culminating in the SPA2D method described in Section IV.

CNN Detection Fails as Target Pixel Count Decreases

We train a YOLO26x object detection CNN [34] on data from the Multi-view Drone Tracking Dataset [35]. For each experiment, we split and resize the data as needed to produce a range of target object sizes. Data are split into training, validation, and test sets under machine learning (ML) best practices. Figure 1 shows a transition from unreliable detection at small pixel counts to reliable detection when targets occupy more pixels. This effect agrees with results from urban cameras [36] and Earth observation [37]–[39]. Thus, CNNs are unsuitable for supporting long-distance rendezvous of ABUs with eVTOLs. An alternative solution is necessary for the long-distance portion of the ABU rendezvous challenge; we expect short-distance operations to make use of ML so long as execution time can be made sufficiently fast.

Shift-Plus-Add Succeeds for Small Pixel Counts

To quantify shift-plus-add performance for distant aerial objects, we modify a MATLAB/Simulink UAV Package Delivery framework with Cesium photorealistic visualization to produce a synthetic dataset with known target positions; we release the simulator and dataset as open source. Figure 2 visualizes simulator output and results for two motion hypotheses. In the middle, Figure 2 shows a correct motion hypothesis that accumulates target pixels coherently as the aircraft moves across the ABU camera view. At the right, Figure 2 shows an incorrect motion hypothesis that fails to accumulate target pixels coherently and instead spreads them across the ABU camera view. All experiments use a Jetson AGX Orin in its 50 W mode to evaluate an edge computing ABU deployment.

Defining the Core Shift-Plus-Add Algorithm

Shift-plus-add makes dim targets observable by accumulating information over time along a motion trajectory. Let $I_t(x, y)$ be the preprocessed pixel value from (x, y) in image I at time t. For a temporal window of N frames and a candidate constant image-plane shift (i.e., pixel-per-frame velocity) hypothesis $\vec{v} = (v_x, v_y)$, accumulate shifted pixels across time:

$$A_{\vec{v}}(x, y) = \sum_{k=0}^{N-1} I_{t+k}\left(x - kv_x, y - kv_y\right), \qquad (1)$$

Sub-pixel values are calculated using linear interpolation.

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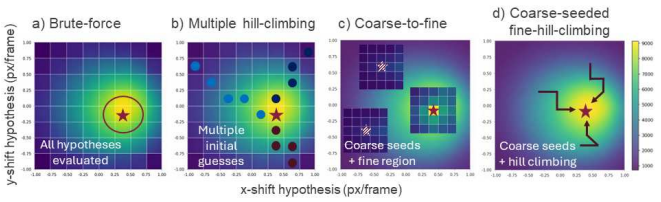


Fig. 3. Illustrations of each motion hypothesis search strategy. Left: Brute-force tests every motion hypothesis. Left-middle: Multiple hill-climbing deploys multiple searchers in parallel. Right-middle: Coarse-to-fine performs detailed searches in parallel after a coarse search. Right: Coarse-seeded fine-hill-climbing deploys multiple searches from the coarse search results.

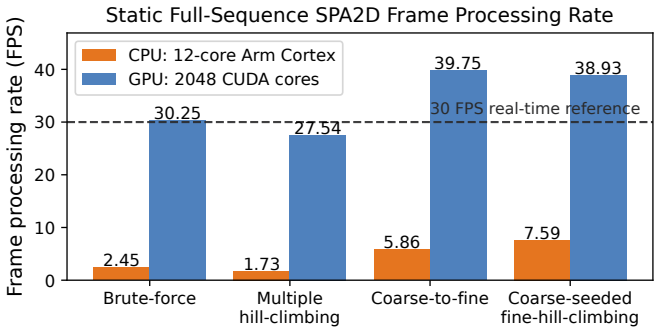


Fig. 4. Frames per second (FPS) for each shift-plus-add hypothesis search method. Brute force provides higher FPS than multiple hill-climbing due to more batching opportunity. Coarse-to-fine provides the highest FPS on GPU.

For each shift hypothesis $\vec{v}$, define its score $S(\vec{v})$ as the maximum value in the accumulated frame:

$$S(\vec{v}) = \max_{x,y} A_{\vec{v}}(x, y) \tag{2}$$

The best shift hypothesis $\vec{v} \in \mathcal{V}$, where $\mathcal{V}$ is the set of candidate pixel-per-frame shift hypotheses, is the hypothesis that gives the maximum score $S(\vec{v})$:

$$\vec{v}_{\text{best}} = \arg\max_{\mathbf{v} \in \mathcal{V}} S(\mathbf{v}), \tag{3}$$

When a shift hypothesis matches the target motion, all target values accumulate in the same (x, y) positions, giving a high score. When a shift hypothesis is incorrect, target values accumulate across different (x, y) positions, giving low scores.

Coarse-to-Fine Supports Real-Time Motion Hypothesis Search

Exploring the set $\mathcal{V}$ of candidate pixel-per-frame shift hypotheses acts as the crux of the shift-plus-add method; a poor search method provides inaccurate hypotheses $\vec{v}$ or search times that lag real-time requirements. This section evaluates four search strategies, motivating the selection of a coarse-to-fine search for the SPA2D method described in Section IV.

Figure 3 illustrates each evaluated search strategy. Brute-force exhaustively scores every motion hypothesis. Multiple hill-climbing performs parallel hill-climbs and returns the highest score. Coarse-to-fine performs a coarse search, followed by local searches seeded from the coarse search. Coarse-seeded fine-hill-climbing uses the same initial coarse search, followed by parallel hill-climbs seeded from the coarse search.

Brute-force search. This strategy evaluates every hypothesis $\vec{v} \in \mathcal{V}$ and selects the one with the highest score $S(\mathbf{v})$. Brute-force always returns the best motion hypothesis and, for a small search space $\mathcal{V}$, completes in real time. As $\mathcal{V}$ grows, brute-force scales poorly in execution time while the quality of the returned motion hypothesis $\vec{v}$ remains constant.

To evaluate brute-force and other searches, we use our representative benchmark (see Figure 2) with a search range of $v_x, v_y \in [-1, 1]$ pixels per frame and a step size of 0.1 pixels per frame. We evaluate hypotheses in batches of 512 deployed to GPU with JAX [40]. Brute-force correctly identifies the true image-plane shift of $(0.7, 0)$ pixels per frame.

Multiple hill-climbing search. This strategy begins with multiple motion hypotheses and evaluates neighboring hypotheses around each candidate. At each iteration, each candidate moves to the neighboring hypothesis with the most improved score; the search terminates when no such neighbor exists (i.e., a local maximum). The search caches visited hypotheses to avoid redundant computation, and all hill-climbs execute in parallel. For benchmarking, we make nine initial guesses in a uniform grid over the search space. Only three of the nine initial guesses converge to the true motion shift vector; the remaining six terminate at local maxima. Figure 4 quantifies the relative execution time of multiple hill-climbing; brute force gives a higher FPS because its regular grid enables more efficient parallel evaluation on both CPU and GPU.

Coarse-to-fine search. Rather than evaluating the full-resolution grid, this method searches a coarse global grid, keeps top coarse candidates, and then evaluates fine local neighborhoods around these candidates. For benchmarking, we use a coarse grid step size of 0.5 pixels per frame, producing 25 coarse hypotheses, and the top five coarse candidates define the fine search neighborhoods. The fine-search stage uses a smaller step size and only evaluates hypotheses near promising coarse responses. This strategy preserves a global search stage while avoiding exhaustive fine-resolution evaluation over the full grid. It converges to the true motion shift vector and, as shown in Figure 4, substantially reduces execution time.

Coarse-seeded fine-hill-climbing search. This strategy combines the coarse global search with hill-climbing. The search first evaluates the same coarse grid, retains top coarse candidates, and then runs parallel hill-climbing from those candidates. Only one of the five coarse-seeded searches converges to the true motion shift vector; the remaining four terminate at local maxima. Figure 4 shows lower GPU FPS, motivating use of the coarse-to-fine search strategy for both the best FPS and the most accurate motion hypothesis.

IV. SPA2D System Design

The motivating studies of Section III show that shift-plus-add detects low-pixel-count targets with motion-compensated temporal accumulation when SotA CNN methods fail. Brute-force trajectory searches are computationally expensive and hill-climbs converge to incorrect local maxima, motivating a coarse-to-fine motion hypothesis search. Given these results, we define the SPA2D detect-and-track system summarized in Figure 5. SPA2D performs target-enhancing preprocessing, windowed shift-plus-add accumulation, coarse-to-fine search, and dynamic local tracking with JAX-based GPU acceleration.

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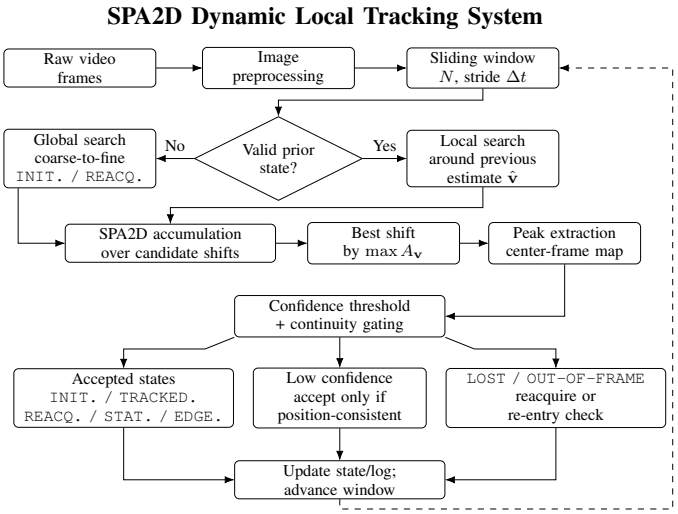


Fig. 5. In SPA2D, raw frames convert to intensity maps for target enhancement. Processing uses overlapping temporal windows. Before initialization (or after target loss) a coarse-to-fine global search is used for acquisition (or reacquisition). After accepting a motion vector, subsequent iterations use a dynamic local search around the previous estimate. SPA2D evaluates each candidate shift, and the top score determines the next iteration.

TABLE I
MULTI-VIEW DRONE TRACKING DATASET [35] FRAME SEQUENCES

Seq.	Frame rate (Hz)	No. of frames	Frame resolution	Target size (pixels)	Preproc. FPS end/core
D1/cam2	50.00	8016	1920×1080	90–400	25.13 / 89.94
D2/cam1	29.84	4749	1920×1080	200–850	20.30 / 46.99
D3/cam3	25.00	14196	1440×1080	40–600	35.37 / 180.74
D4/cam3	21.24	10933	3840×2160	40–4500	10.66 / 33.73

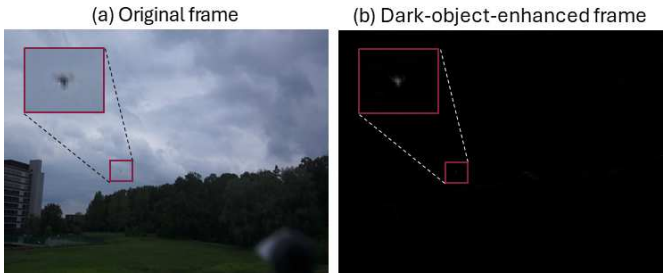


Fig. 6. A D3/cam3 image frame before and after target-enhancing preprocessing. The tracked object is a hexacopter UAV. Left: Original frame. Right: Dark-object-enhanced image used by SPA2D. Insets magnify target region.

System Overview

The SPA2D detect-and-track loop processes a video from an ABU as a sequence of overlapping temporal windows. Raw frames first convert to intensity images for target enhancement. A temporal window $\mathcal{W}_t$ beginning at frame I_t contains N consecutive, preprocessed frames:

$$\mathcal{W}_t = \{I_t, I_{t+1}, \ldots, I_{t+N-1}\}. \tag{4}$$

At each processing step, the window start index advances by a stride of Δt frames. SPA2D reports the target state at the center frame of each window:

$$\mathbf{s}_{c(t)} = \begin{bmatrix} x_{c(t)} & y_{c(t)} & v_{x,t} & v_{y,t} \end{bmatrix}^T, \tag{5}$$

where $(x_{c(t)}, y_{c(t)})$ represent the image-plane position and $(v_{x,t}, v_{y,t})$ represent the pixel-per-frame velocity.

As shown in Figure 5, SPA2D has two search modes. Before initialization (or after target loss) SPA2D performs a coarse-to-fine global search over the configured velocity bounds for acquisition (or reacquisition). During tracking, the search restricts to a neighborhood around the previously tracked target motion. Both search modes feed data to the same accumulation and peak-extraction module; only the candidate shift set changes. This strategy preserves the robustness of global acquisition while reducing the number of hypotheses evaluated during continuous tracking (i.e., the common case).

Image Preprocessing

The preprocessing step converts RGB frames to target-enhanced intensity images. SPA2D accumulates images along candidate trajectories, so the input representation affects the resulting score. A representation with stronger separation between target and background pixels produces larger accumulated values for correct velocity hypotheses, while poor separation creates competing peaks along incorrect hypotheses.

This step produces a dark-object-enhanced image. Let G_t be the grayscale image at frame t, and let $\texttt{close}(G_t, \mathcal{B})$ be a morphological closing operation — i.e., a classical computer vision function composed of a dilation followed by an erosion to eliminate small and narrow image features — with an elliptical kernel $\mathcal{B}$. Subtracting the grayscale image from its morphological closing gives the dark-object-enhanced image:

$$D_t = \texttt{close}(G_t, \mathcal{B}) - G_t, \tag{6}$$

In practice, we call OpenCV's morphological closing function.

Windowed Shift-Plus-Add Tracking

During a temporal window, SPA2D approximates target image-plane velocity as constant. As listed in Section III, the motion-compensated velocity hypothesis accumulation is:

$$A_{\vec{v}}(x, y) = \sum_{k=0}^{N-1} I_{t+k}\left(x - kv_x, y - kv_y\right), \tag{7}$$

Also as listed in Section III, score and shift hypothesis are:

$$S(\vec{v}) = \max_{x,y} A_{\vec{v}}(x, y) \qquad \vec{v}_{\text{best}} = \arg\max_{\mathbf{v}\in\mathcal{V}} S(\mathbf{v}), \tag{8}$$

Given $\vec{v}_{\text{best}}$, SPA2D estimates target location as the accumulated response peak. This peak represents the target position in the first frame of the window. To advance the target state to the temporal center of the window, position is calculated as:

$$\vec{p}_{c(t)} = \vec{p}_{0,t} + \frac{N-1}{2}\vec{v}_{\text{best}} \tag{9}$$

SPA2D reports target position at frame $c(t)$ to reduce temporal bias when consecutive windows overlap.

Window length N gives a tradeoff between accumulation and motion-model fidelity. Longer windows provide stronger temporal integration for faint targets but are less tolerant of acceleration or maneuvering. Shorter windows are more responsive to changing motion but provide weaker accumulation, opening the potential of lower performance for low-SNR targets.

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TABLE II
SPA2D TRACKING PERFORMANCE FOR EACH FRAME SEQUENCE. D1/CAM2 AND D2/CAM1 REPRESENT ABU-LIKE SKY-TRACKING CASES, AND D3/CAM3 AND D4/CAM3 ACT AS STRESS TESTS. FRAME PROCESSING RATE IS MEASURED IN FPS ON THE JETSON AGX ORIN 64GB IN ITS 50 W MODE.

Sequence	Role	Window overlap	Global $(v_{\max}, \delta_c)$	Fine (r_f, δ_f)	Local (r_l, δ_l)	Window length N	Stride	FPS	Valid tracking outcome (%)	Recall@ 100px (%)	Precision@ 100px (%)	RMSE$_p$ (px)	RMSE$_v$ (px/frame)
D1/cam2	ABU-like	No	$(1, 0.5)$	$(0.2, 0.1)$	$(0.3, 0.1)$	5	5	**27.91**	99.50	**100.00**	**98.99**	**4.43**	0.26
D1/cam2	ABU-like	Yes	$(1, 0.5)$	$(0.2, 0.1)$	$(0.3, 0.1)$	5	1	9.49	99.41	100.00	98.82	4.48	1.22
D2/cam1	ABU-like	No	$(4, 0.5)$	$(0.2, 0.1)$	$(0.3, 0.1)$	5	5	**16.53**	97.79	**97.38**	**99.55**	12.43	1.42
D2/cam1	ABU-like	Yes	$(5, 0.5)$	$(0.2, 0.1)$	$(0.3, 0.1)$	5	1	5.36	98.69	99.21	98.98	11.64	4.65
D3/cam3	Stress-test	Yes	$(15, 2.0)$	$(0.4, 0.2)$	$(0.6, 0.25)$	8	4	10.57	93.86	92.41	93.72	4.28	0.58
D4/cam3	Stress-test	Yes	$(15, 2.0)$	$(1.0, 0.2)$	$(1.0, 0.25)$	8	4	1.93	85.84	84.64	93.36	23.11	5.69

Initialization and Reacquisition

Initialization uses coarse-to-fine velocity search (see Section III). The coarse stage evaluates a sparse global grid over the configured velocity range and retains top-scoring hypotheses that then define fine local neighborhoods $\mathcal{N}$ with a smaller velocity search step. The union of these neighborhoods forms the initialization candidate set $\mathcal{V}$:

$$\mathcal{V}_t^{\text{init}} = \bigcup_{\vec{v}_i \in \mathcal{C}_t} \mathcal{N}(\vec{v}_i), \tag{10}$$

where $\mathcal{C}_t$ is the set of retained coarse candidates and $\mathcal{N}(\vec{v}_i)$ is the fine neighborhood around candidate $\vec{v}_i$. This same search is used for reacquisition when the SPA2D loses the target.

Dynamic Local Tracking

After beginning to track a target, SPA2D transitions from acquisition to dynamic local search. Rather than evaluating the full initialization set, the candidate velocity set centers around the previously accepted velocity:

$$\mathcal{V}_t^{\text{local}} = \left\{ \vec{v}_{t-\Delta t} + \begin{bmatrix} a\delta_l \\ b\delta_l \end{bmatrix} : a, b \in \mathbb{Z},\ |a\delta_l| \le r_x,\ |b\delta_l| \le r_y \right\}, \tag{11}$$

where δ_l is the velocity step and r_x and r_y are local search radii. Accumulation and scoring over $\mathcal{V}_t^{\text{local}}$ are unchanged.

The dynamic local search succeeds when the image-plane velocity changes slowly, as is the case for ABU tracking of eVTOL targets. If dynamic local tracking loses the target, SPA2D returns to coarse-to-fine search for reacquisition.

V. METHODOLOGY

We implement SPA2D in Python using JAX [40] for GPU acceleration of motion hypothesis testing. SPA2D batches candidate velocities and vectorizes accumulations across hypotheses. As a result, SPA2D evaluates many motion hypotheses simultaneously on the GPU. SPA2D uses the same batched accumulation for both initialization and local tracking; only the candidate set $\mathcal{V}_t$ changes between search modes.

The implementation separates hypothesis scoring from target localization. During search, SPA2D only calculates the scalar score $S(\vec{v})$ for each velocity. After selecting $\vec{v}_{\text{best}}$, the corresponding accumulated response estimates the target peak. This formulation reduces unnecessary computation during hypothesis evaluation while preserving accurate image-plane localization for the accepted trajectory. SPA2D records per-window timing and effective tracking rate, enabling the system to be evaluated as a real-time object detection and aerial perception module.

Experimental Setup

We select videos from the Multi-view Drone Tracking Dataset [35], which captures a flying hexacopter UAV from multiple ground-based cameras. The selected sequences vary in frame rate, frame count, and resolution; background clutter; and target pixel count, velocity, and acceleration. "Dataset1/cam2" (D1/cam2) and "Dataset2/cam1" (D2/cam1) represent ABU-like sky-tracking cases because they contain smoother image-plane motion and moderate-resolution images. "Dataset3/cam3" (D3/cam3) and "Dataset4/cam3" (D4/cam3) represent stress tests with faster motion, higher acceleration, higher resolution, and challenging backgrounds. Figure 6 shows a representative D3/cam3 frame and the corresponding target-enhanced image used by SPA2D. In all cases, we apply the same SPA2D procedure described in Section IV. Table I summarizes the selected datasets.

During evaluation, we fix the accumulator thresholds for D1/cam2 and D2/cam1 to illustrate that additional sequence-specific tuning can further improve performance. We report three sequence-level tracking metrics — visible-target recall, precision, and valid tracking outcome. Visible-target recall at $100\,\text{px}$ counts a visible target as successfully recovered when the Euclidean distance between the estimated image-plane position and the ground-truth center is within $100\,\text{px}$. Precision at the same $100\,\text{px}$ criterion measures the fraction of accepted target estimates that are correct; estimates during target absence or farther than $100\,\text{px}$ from a visible target count as false positives. Valid tracking outcome measures whether SPA2D produces a valid state over the sequence, including correct visible-target estimates, valid target-absent states, and out-of-frame handling. We compute position and velocity root mean square error (RMSE) using the Euclidean errors of the two-dimensional position and velocity vectors, respectively. All execution times are measured on the NVIDIA Jetson AGX Orin 64GB in its $50\,\text{W}$ mode.

VI. EVALUATION AND RESULTS

Results show that SPA2D provides accurate tracking of ABU-like scenarios in real time onboard an embedded GPU. Stress tests (less-cooperative targets) are also tracked accurately. SotA CNNs struggle in this small-target regime; SPA2D addresses this capability gap.

SPA2D Provides Accurate Tracking of Distant Targets

Table II reports SPA2D tracking accuracy. The global search column denotes the symmetric coarse shift range and step size, $(v_{\max}, \delta_c)$, corresponding to $v_x, v_y \in [-v_{\max}, v_{\max}]$ for both

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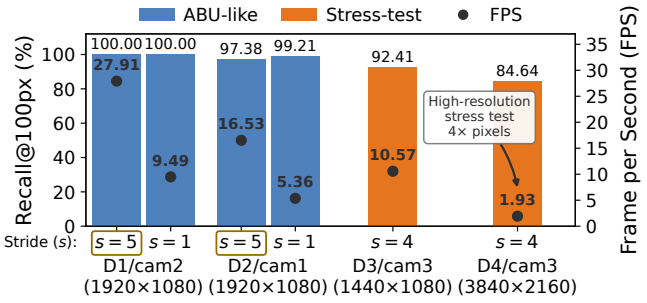


Fig. 7. SPA2D accuracy and FPS. Bars show the recall score described in Section V, and black markers indicate FPS from Table II. The ABU-like cases maintain high recall and real-time edge processing for autonomous flight.

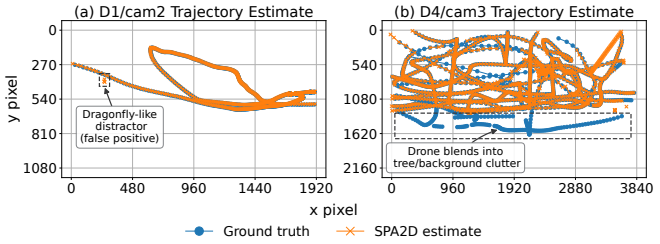


Fig. 8. (a) D1/cam2 shows an ABU-like sky-tracking case where the returned trajectory closely follows the true trajectory; the false positive is due to a detected insect. (b) D4/cam3 shows the most challenging stress test with aggressive motion, high resolution, and clutter causing missed intervals.

axes. The fine-search column reports the symmetric refinement radius and step size, (r_f, δ_f), and the local-search column reports the symmetric local radius and step size, (r_l, δ_l).

In ABU-like sky-tracking, non-overlapping windows give better tradeoffs between accuracy and execution time. For D1/cam2, window size $N = 5$ and stride 5 achieves 28 FPS at 100% visible-target recall, 99% precision, and $4.4\,\mathrm{px}$ position RMSE. A stride of 1 lowers frame processing rate to 9.5 FPS without improving recall or position RMSE. D2/cam1 follows the same pattern: stride 5 achieves 17 FPS, 97% recall, and 99.6% precision, while stride 1 increases recall to 99% but reduces frame rate to 5 FPS.

Stress tests D3/cam3 and D4/cam3 require wider search ranges and window overlap because target motion is faster, less regular, and affected by background clutter. With $N = 8$ and stride 4, D3/cam3 achieves 92% recall, 94% precision, and $4.3\,\mathrm{px}$ position RMSE at 11 FPS. The high-resolution stress test D4/cam3 is the most difficult case because its 3840×2160 resolution greatly increases the cost of full-frame accumulation, and the target repeatedly exits, re-enters, or blends into background clutter — triggering more frequent (computationally costly) reacquisition states. SPA2D maintains 85% recall and 93% precision in this case, but frame processing rate drops to 2 FPS. Thus, high recall is not achieved by indiscriminately returning target estimates: precision remains above 98% for the ABU-like cases and above 93% for both stress tests. In the ABU application, the target eVTOL aircraft acts cooperatively with the ABU; D4/cam3 represents an adversarial lower bound.

Constrained Search Balances Accuracy and Frame Rate

The global-search sweep further shows the value of constraining the motion hypothesis space. For the first ABU-like case, expanding the global search range from ± 1 to ± 5 pixels per frame leaves recall at 100% but reduces frame rate from 28 to 19 FPS. For the second ABU-like case, expanding the range from ± 1 to ± 4 pixels per frame increases recall from 90% to 97%, while frame rate slightly decreases from 19 to 17 FPS; increasing the range further does not improve recall. These results favor the smallest search range that captures the expected target motion and support using cooperative flight information to constrain SPA2D searches.

Figure 7 shows recall and frame rate for each application. D1/cam2 and D2/cam1 both achieve high recall, while D2/cam1 requires a wider global search range that lowers FPS even with non-overlapping windows. The ABU-like cases illustrate similar tracking performance under different search-space requirements. Stress test sequences reduce frame rate because they require wider search, more overlap, higher-resolution processing, or more frequent reacquisition.

Figure 8 visualizes the trajectories returned by SPA2D. The contrast between D1/cam2 and D4/cam3 matches the results in Table II: SPA2D maintains close trajectory agreement in the ABU-like sky-tracking case, while the cluttered, high-resolution stress test produces more missed intervals.

SPA2D Provides Tracking in Real Time

SPA2D frame processing rates reflect the effects of window size, search space, image resolution, and reacquisition frequency. Moderate-resolution 1920×1080 ABU-like frames with non-overlapping windows reach 28 FPS on D1/cam2 and 17 FPS on D2/cam1 while maintaining high recall. Overlapping windows reduce frame rate, and a wider global search increases motion candidate count — explaining why D2/cam1 runs more slowly than D1/cam2 despite comparable tracking performance. The stress tests require wider search and window overlap. D4/cam3 is especially costly because each 3840×2160 frame contains $4\times$ as many pixels as a 1920×1080 frame, requiring more full-frame accumulation time per hypothesis. Repeated low-confidence or out-of-frame states trigger reacquisition searches, further reducing FPS.

Preprocessing measurements in Table I show that the target-enhancement stage is not the primary execution time bottleneck when image I/O and data transfer are excluded. In an onboard, edge-computing, real-time implementation, frames stream directly to the tracker rather than being written as intermediate images. Results indicate that SPA2D performs best for smaller image resolutions and a constrained motion candidate search space. This work evaluates vision-only searches without data fusion; an ABU makes use of additional information such as ADS-B, flight plans, air traffic notices, and relative velocity and heading hints to restrict the motion search space. ABUs contain sufficient size, weight, and power, and cost (SWaPC) to support more performant desktop-class GPUs compared to the embedded GPU of a Jetson Orin. Results support the conclusion that motion-compensated temporal accumulation via shift-plus-add tracks and detects low-pixel-count aerial targets that are dim in individual frames, addressing a known

gap in SotA object detection with ML while applying shift-plus-add processing techniques to the ABU aerial regime.

VII. Related Work

Vision-Based Sense-and-Avoid and Target Detection

Unmanned aircraft system (UAS) research consistently investigates vision-based sense-and-avoid and target detection. Surveys highlight challenges in long-range obstacle detection, false alarms, background clutter, and real-time onboard perception [4]. Some approaches use staged processing pipelines [2] and motion cues [3]. Subsequent work addresses passive monocular ranging [5], below-horizon detection [6], [7], learning-based classification [8], and onboard detection and tracking [9], [10]. Drone detection studies consider small aerial targets [12] and broader anti-UAV challenges [13]. In contrast to these detection-first systems, SPA2D directly accumulates image data before detection and tracking.

Small-Object Detection in Aerial Imagery

Small-object detection in aerial images advances through multi-scale features, high-resolution detection heads, lightweight networks, and attention-based feature fusion. Foundational object detection architectures, including region-based convolutional networks [14], YOLO [15], feature pyramid networks [16], DETR [17], and shifted-window transformers [18], provide the basis for many aerial detection systems. Recent UAV and drone-view detection methods improve small-object performance with cross-ensemble feature fusion [20] and cross-scale transformer attention [19]. Tracking-oriented aerial work addresses drone-based multi-object tracking [41] and scale-aware adaptation for UAV tracking [42]. These methods struggle with far-range rendezvous targets that occupy few pixels and make no real-time claims for autonomous edge computing.

Infrared and Dim-Small-Target Detection

Infrared small-target detection addresses a related low-SNR perception challenge where targets often lack texture, shape, or clear boundaries. A recent survey summarizes classical and learning-based approaches for single-frame and dim-small-target detection [24]. Specific methods use adversarial segmentation [43], attentional local contrast [23], dense nested attention [22], and scale- and location-sensitive learning [21]. Thermal sensing combined with ADS-B [44] and fused acoustical-optical detection [45] provide additional multimodal approaches. SPA2D shares the goal of recovering weak targets, but differs by focusing on visible-image aerial rendezvous and by using motion-compensated temporal accumulation rather than modality-specific infrared target-background models.

Temporal Accumulation of Sensor Data

Temporal information supports cooperative visual tracking [11], real-time aerial localization and tracking [46]. Track-before-detect methods have also explored temporal evidence accumulation for weak targets. One approach evaluates sampled motion hypotheses using accumulated target likelihoods, rather than directly shift-and-adding image data [47]. Coarse-to-fine particle-filter search has also been applied to long-range, low-SNR targets [48]; its similarity to SPA2D is limited to the use of hierarchical search. Astronomical synthetic tracking is closer to the core shift-and-add operation, searching candidate velocity vectors and accumulating motion-compensated frames [33], with GPU-parallel implementations also demonstrated on small-telescope systems [49]. SPA2D differs from these approaches by applying image-plane shift-plus-add to long-range ABU rendezvous and demonstrating the method on embedded hardware.

VIII. Conclusion and Future Work

This work presents SPA2D, a shift-plus-add track-before-detect framework for Autonomous Battery Units (ABUs). In an ABU rendezvous, a distant host aircraft (e.g., an eVTOL) occupies only a small number of pixels and lacks reliable single-frame features. SPA2D searches over image-plane shift hypotheses (i.e., candidate target trajectories) and accumulates motion-compensated image data. Experiments show that SPA2D maintains tracking continuity on ABU-like sky-tracking cases, achieving 100% visible-target recall on the D1/cam2 dataset and between 97% and 99% recall on the D2/cam1 dataset. SPA2D remains effective on more aggressive stress tests, including 92% recall on the D3/cam3 dataset and 85% recall on the high-resolution D4/cam3 dataset. The strong recall performance is achieved without sacrificing precision, which stays above 98% for the ABU-like cases and above 93% across both stress tests.

Future work will incorporate additional target data available to ABUs, such as expected approach geometry, traffic information, GPS, and host aircraft trajectory, to guide region-of-interest search and improve frame rate. Additional directions of research include multi-target tracking and three-dimensional trajectory estimation by combining image-plane motion with known target size, apparent pixel scale, ABU position, camera characteristics, and cooperative navigation information. Future evaluation will develop and utilize additional datasets to better characterize false alarms and assess scenario generalization.

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